RAJALAKSHMI ENGINEERING COLLEGE	**RAJALAKSHMI ENGINEERING COLLEGE** **B.E. COMPUTER SCIENCE AND ENGINEERING** **(CYBER SECURITY)** **CR23721 - PROJECT ABSTRACT (2026-27)**

TITLE	Quantum Resistant Blockchain Security Framework
TEAM MEMBERS **REG. NUM &NAME**	231901008 - DHARSHNA U
PROJECT GUIDE	Dr. Tamilselvi P. (HOD CSECS)
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ABSTRACT:

The rapid advancement of quantum computing poses a significant challenge to the long-term security of blockchain-based healthcare systems, where sensitive patient and pharmaceutical data require both confidentiality and integrity. Existing solutions primarily rely on classical cryptographic algorithms and reactive security mechanisms, making them vulnerable to future quantum threats and sophisticated cyberattacks. **QuantumShield** presents a secure **permissioned blockchain framework** that combines **Hyperledger Fabric**, NIST-standardized **Post-Quantum Cryptography** (**ML-KEM and ML-DSA**), and **AI-driven anomaly detection** to provide proactive and **quantum-resilient security for healthcare and pharmaceutical collaboration**. The framework analyses transaction behaviour to identify suspicious activities before they are committed to the blockchain, while a validator reputation mechanism strengthens trust among participating organizations. A real-time security dashboard and performance benchmarking further enhance system visibility and evaluate the impact of post-quantum cryptography on blockchain performance. By integrating quantum-safe cryptography with intelligent threat detection in a unified architecture, QuantumShield delivers a **scalable** and **future-ready platform** for secure digital collaboration in healthcare ecosystems.

PROJECT GUIDE **PROJECT COORDINATOR**